

VITISH 2026

(SIH Internal Hackathon)

The design file already knows what should be on the board.

Problem Statement ID: VITISH-82

Problem Statement Title: Low-cost Optical Inspection for PCB Assembly

Theme: Smart Automation / Manufacturing

PS Category: Hardware

Idea Title: GerberEye — CAD-Referenced Optical Inspection

Team Name: <team name>

Team ID: <team id>

154

unit tests passing

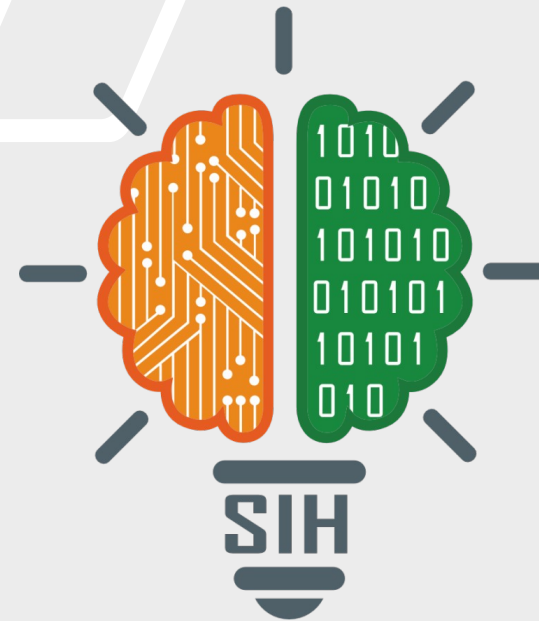
29/29

integration checks

0

network calls made

Commercial AOI: ₹15–50 lakh. This: **under ₹10,000**, on the laptop the shop already owns.



IDEA TITLE — GerberEye: CAD-Referenced PCB Inspection

THE PROBLEM

- ▶ An MSME inspects assembled PCBs **by eye**, under a magnifier lamp
- ▶ ~**200 reference designators** cross-checked against a printed drawing
- ▶ Accuracy falls with **fatigue and board density**; nothing is recorded
- ▶ Automating it costs **₹15–50 lakh** — more than the shop earns in a year

THE INSIGHT

- ▶ The shop **already owns the design files** — it cannot manufacture without them
- ▶ The **pick-and-place file** states what sits at every coordinate
- ▶ So: **the design file is the label**
- ▶ **No training data. No annotation. No dataset needed.**

HOW IT WORKS

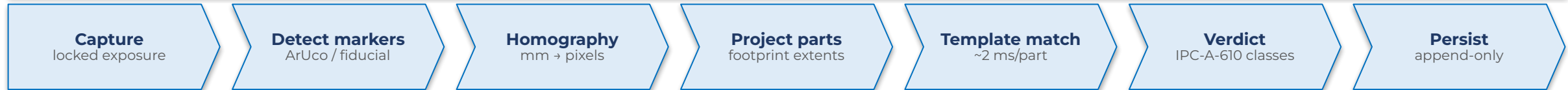


INNOVATION & UNIQUENESS

	Conventional approach	GerberEye
Reference	Operator memory / printed drawing	The CAD file itself
Training data	Thousands of labelled defect images	None — the P&P file is the label
Output	"This board looks wrong"	"U1 rotated 90°, C2 missing"
Cost / connectivity	₹15–50 lakh, vendor-tied	< ₹10,000, fully offline

TECHNICAL APPROACH

INSPECTION PIPELINE — CPU only, no GPU, ~100 ms per board



TWO-PATH ARCHITECTURE (ADR-002)

Path	Needs	Gives
A — Golden differencing	One known-good board	Regions that differ
B — CAD registration	Gerber + pick-and-place	Defects named by designator

Path A works with **zero design files**, so the demo cannot be blocked by a missing CAD↔board pair. Path B is the differentiator.

TECHNOLOGY STACK — permissive licences only

Layer	Choice	Licence
Computer vision	OpenCV	Apache-2.0
CAD parsing	pygerber	MIT
Backend	Python · FastAPI · SQLite	MIT
Operator UI	React + Vite	MIT
Compute	Laptop CPU / Raspberry Pi	—

Why not YOLO / deep learning? The latency budget allows ~6 ms per component on a CPU with no GPU — and Ultralytics is AGPL-3.0, so an MSME deploying it would inherit a copyleft obligation. Every dependency here is MIT, Apache-2.0 or BSD.

The hard part is registration, not comparison. CAD is in millimetres about a board datum; the camera sees pixels at unknown scale and perspective. Solving that homography is what makes "is C2 present?" answerable — and it needs Gerber knowledge **and** computer vision.

SOLUTION FEASIBILITY & VIABILITY

ALREADY BUILT AND VERIFIED

Capability	Evidence
Full pipeline, end to end	154 unit tests, 0 failing
Frontend ↔ backend wiring	29 / 29 integration checks
Defect naming + classification	absent / rotated / offset + measurement
Audit trail	append-only records, CSV export
Runs with no camera at all	demo mode + simulated bench

INDICATIVE BILL OF MATERIALS

Item	Cost
USB webcam, 1080p	₹1,200
LED ring light + diffuser	₹900
Fixed jig, mount, enclosure	₹1,600
Raspberry Pi (optional edge node)	₹4,800
Laptop	already owned
Total	< ₹10,000

RISKS AND HOW EACH IS CONTAINED

Risk	Why it matters	Containment
Illumination instability	Dominant term in the false-call rate — larger than any algorithm change	Stability gate refuses to trust thresholds above 2 grey levels; degraded state shown, not hidden
Camera refuses exposure lock	Breaks golden-board comparison between captures	Detected and reported; enclosure controls ambient light; reproduced in simulation for testing
No matched CAD ↔ board pair yet	Path B cannot be demonstrated on that board	Two-path design — Path A needs no design files at all
Hardware failure on demo day	Loses the live demonstration	Demo mode is a complete working station with no camera attached

Stated honestly: recall and false-call rate are **not yet measured** — that needs seeded defects on real boards. We offer falsifiability instead: hand us a board, pull a component, and we will re-run it in front of you.

REAL WORLD IMPACTS, BENEFITS & RELEVANCE

WHAT CHANGES ON THE SHOP FLOOR

	Today	With GerberEye
Method	Eye + magnifier + printed assembly drawing	Camera + the shop's own design files
Time per board	2–5 minutes of concentrated attention	Seconds, unattended
What the operator is told	"Something looks wrong here"	"U1 rotated 90°. C2 missing. R5 off its pads by 1.6 mm"
Record of inspection	A signature on a sheet	Machine-generated record, exportable as CSV
Cost to automate	₹15–50 lakh	Under ₹10,000

ECONOMIC

- ▶ Defects caught **in-house**, not at the customer's incoming inspection
- ▶ An inspection trail lets an MSME **bid for contracts** that require one
- ▶ Recurring-defect trends turn detection into **process improvement** — a feeder fault, not a rework job

SOCIAL

- ▶ **The operator is always the authority** — one-click override, no form to fill
- ▶ Augments the inspector rather than replacing them; **no de-skilling**
- ▶ Removes the fatigue-driven error that eye inspection makes inevitable

STRATEGIC

- ▶ **Frugal Industry 4.0** for MSMEs — aligned with SAMARTH Udyog Bharat 4.0
- ▶ **Zero network egress** — customer design IP never leaves the machine
- ▶ Built to **IPC-A-610** defect taxonomy; **IPC-2591 (CFX)** ready for MES integration

CAD software checks whether the design is correct. GerberEye checks whether the board in your hand matches it. Design defects happen once; assembly defects happen on every unit, on every shift.

TEAM CAPABILITY

TEAM AND ROLES — 5 CSE + 1 ECE

Stream	Ownership
Capture & golden reference	ECE — jig, lighting, camera settings
Differencing engine (Path A)	CSE — computer vision
CAD registration (Path B)	CSE — Gerber / pick-and-place, homography
Backend & data	CSE — FastAPI, SQLite, records
Operator UI	CSE — React, overlay, override
Demo, risk & pitch	Team lead

PROOF OF WORK — this repository

Artefact	Scale
Requirements engineering	27 functional, 13 non-functional, 16 stakeholder
Architecture	6 ADRs, C4 model, ATAM-lite evaluation
Implementation	~7,800 lines; 154 tests; 29 integration checks
Traceability	Every requirement traced both directions, orphan-checked
Working prototype	Runs end to end today, with no hardware

WHY THIS DOMAIN — AND WHAT WE FOUND IN OUR OWN SYSTEM

We benchmarked against a **221-component 0603 board** rather than our own demo board. **Two of five seeded defects were missed — both of them 'missing components'**, the class that should be easiest. A removed 0603 changes ~100 px² of contour area; the default noise floor was 120 px², so the region was discarded and **the board reported PASS**. That is an escape, not a false call — the dangerous direction. The station now computes the smallest component's expected change from its footprint and **warns when the threshold would hide it**.

Finding a silent recall failure in your own system, and building the guard that surfaces it, is the capability we would most like to be judged on.

RESEARCH AND REFERENCES

STANDARDS AND SPECIFICATIONS

Reference	What it gives the project
IPC-A-610 — Acceptability of Electronic Assemblies	The defect taxonomy: absent, misaligned, rotated, polarity-reversed, tombstoned
IPC-2591 (CFX) — Connected Factory Exchange	The MES integration path — how an inspection station reports to a factory line
Gerber RS-274X / X2 (Ucamco specification)	Board geometry; X2 attributes carry component and net metadata
Pick-and-place / centroid file (CAD output)	Reference designator, X, Y, rotation, side — the ground truth used as the label

LIBRARIES AND TECHNIQUES

- **OpenCV** — ArUco marker detection, homography (RANSAC), ECC alignment, template matching
- **pygerber** — Gerber rendering to raster for CAD-to-camera reference
- **Golden-board differencing** — the classical AOI baseline, used as the no-design-files path
- Surveyed **PatchCore / anomalib** few-shot anomaly detection — deferred: CPU latency budget

FINDINGS THAT SHAPED THE DESIGN

- Public PCB datasets (DeepPCB, HRIPCB) are **bare-board trace defects** — the wrong problem entirely
- **No usable public dataset of PCB assembly defects exists** → so the design file must be the label
- GitHub survey: **no prior art** for projecting Gerber/pick-and-place onto a live camera frame
- Commercial AOI is vendor-locked and cloud-tied — **offline operation is a requirement, not a feature**

Working prototype, requirements set and architecture decision records: github.com/Haise-727/pcb-catcher